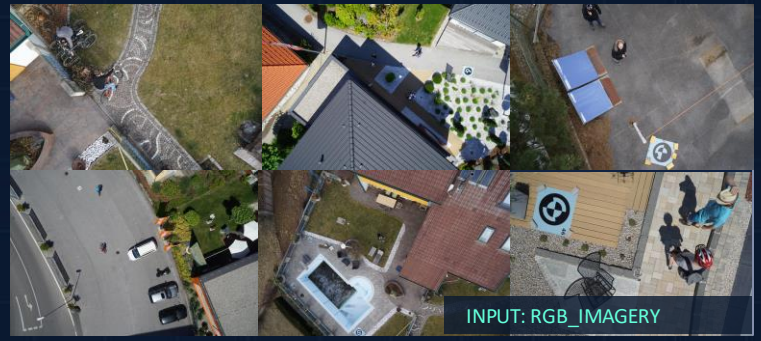
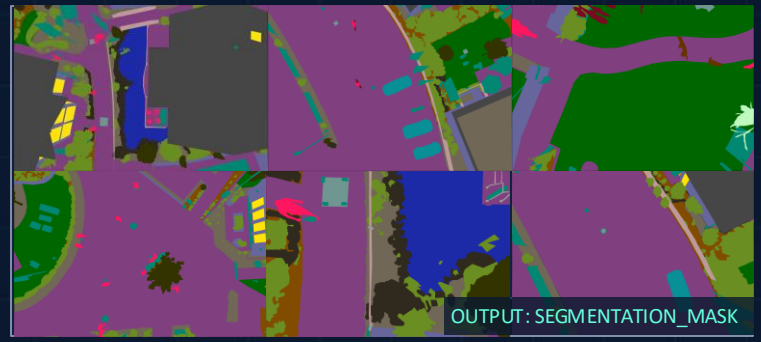
Aerial Semantic Segmentation Using Deep Learning

Vegetation Detection from Drone Imagery

Computer Vision Project
Fernandez, Mecchia, Nicolet

- Patches binary classification
- Semantic Segmentation



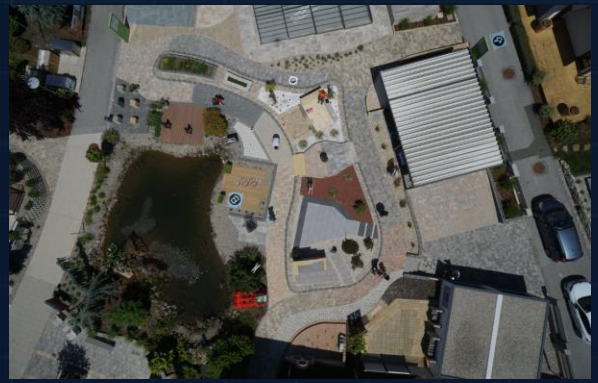


Dataset Overview

Aerial Semantic Segmentation Drone Dataset

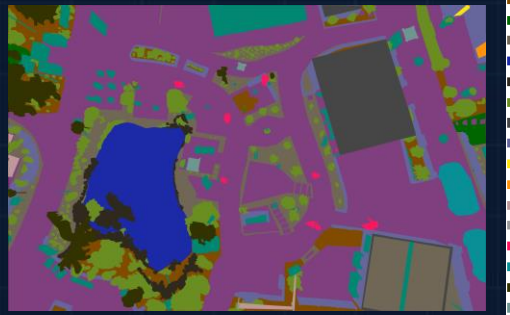
DATASET CHARACTERISTICS

- Images captured from UAVs
- Total of 400 different images
- Total of 24 different labels
- High resolution of 6000x4000 pixel
- class_dict.csv containing RGB to class mapping



Label-Image-Semantic

RGB-color-image-masks



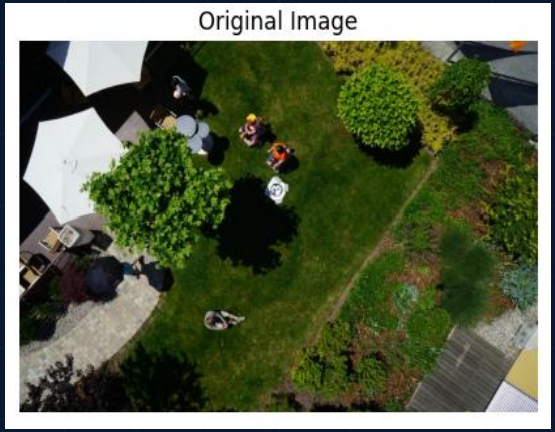
Patches binary classification

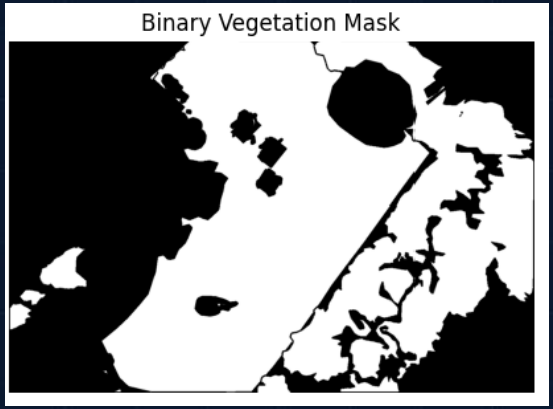
Binary Mask Generation

Unified Vegetation Class Strategy

Classes remapped via CSV to focus on vegetation:

ORIGINAL CLASS	TARGET CLASS	TYPE
VEGETATION GRASS	VEGETATION	BINARY (1)
OTHERS	BACKGROUND	BINARY (0)





Patch Extraction Strategy

Stratified Random Sampling for Balanced Training

SAMPLING LOGIC

To ensure robust model learning, we extract fixed-size patches from each source image based on the binary mask.

- **3 Random Points** from Vegetation
- **3 Random Points** from Non-Vegetation
- **32x32 patches sizes**

2,376
Total Patches Extracted

1,188
Vegetation (50%)

1,188
Non-Veg (50%)

SOURCE IMAGE → RANDOM SAMPLING → CROP PATCHES → BALANCED DATASET

Model Comparison

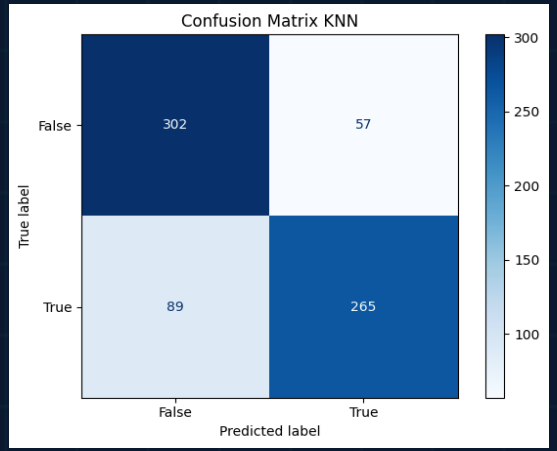
Two Approaches for Vegetation Detection

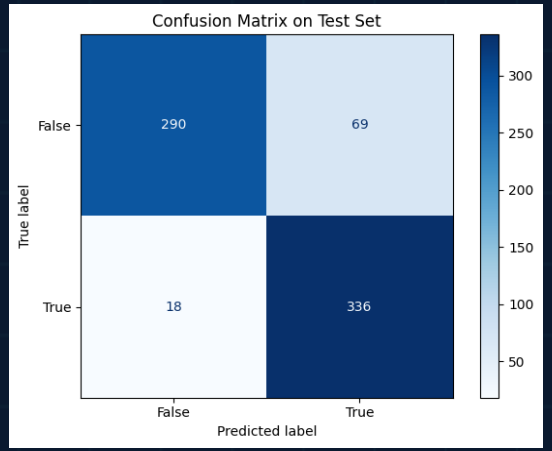
KNN (Baseline)

- Classifies based on feature similarity (color/texture) to nearest neighbors
- Simple, interpretable, zero training time
- Struggles with complex spatial patterns
- **Accuracy: 77.52%**

CNN (Deep Learning)

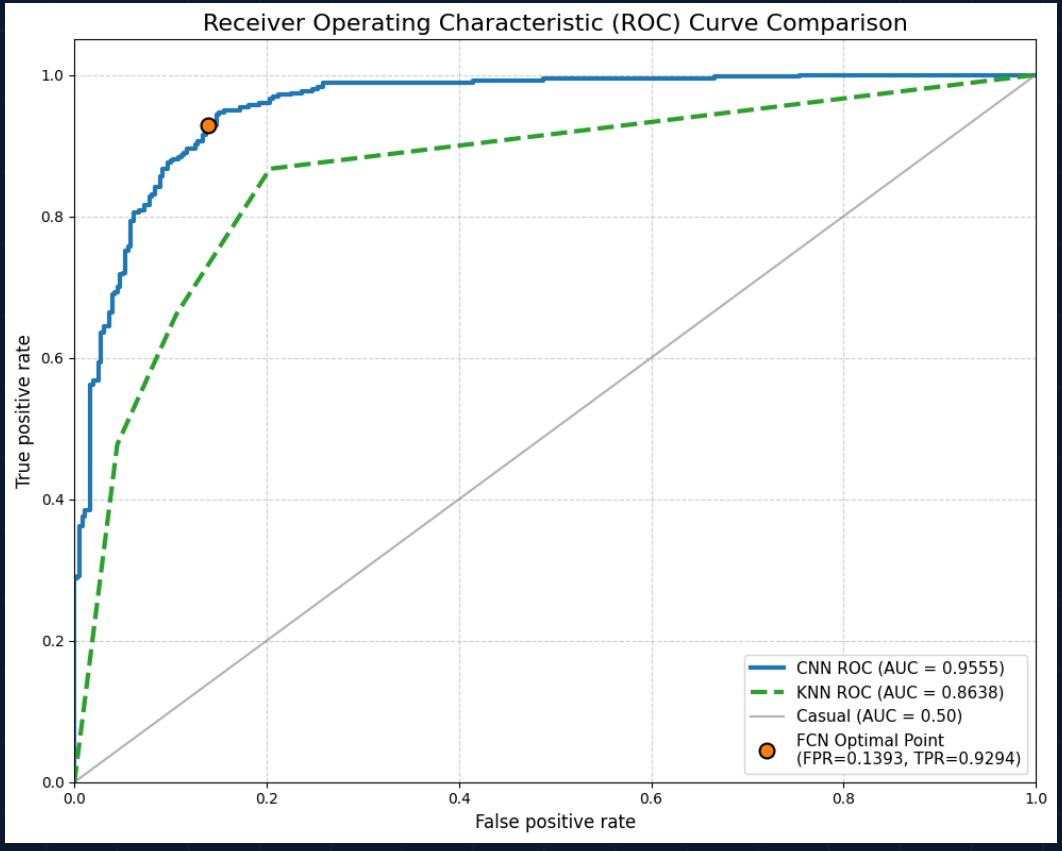
- Learns hierarchical spatial features via backpropagation
- Advanced classifier for complex pattern recognition
- Captures spatial hierarchies, robust to variations
- **Accuracy: 88.73%**





Model Comparison

ROC Curve



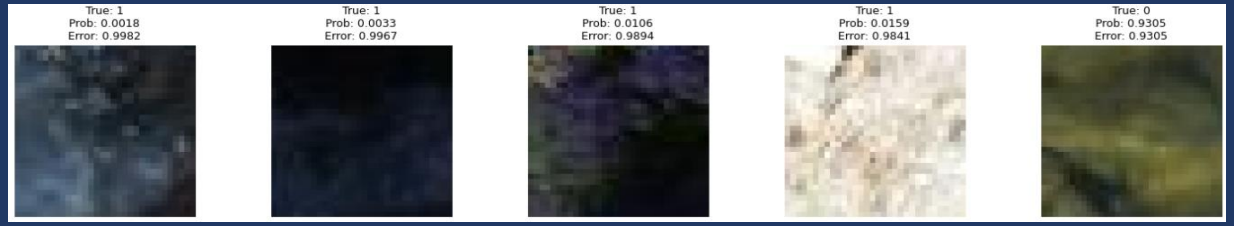
CNN Classification Analysis

Most Confident patches – Highest Errors

Lowest Probability

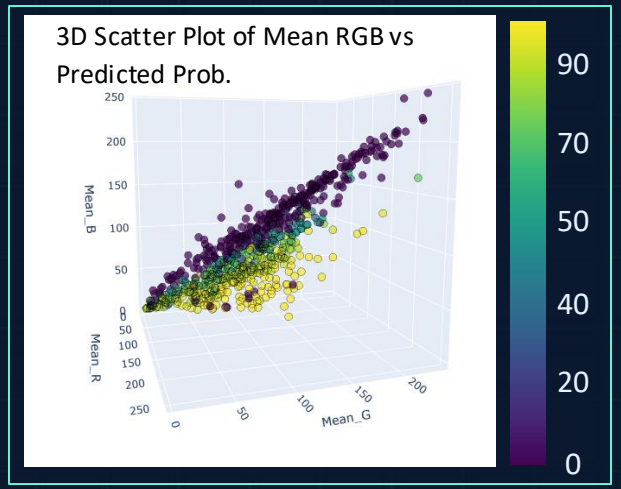
Highest Probability

Highest Error

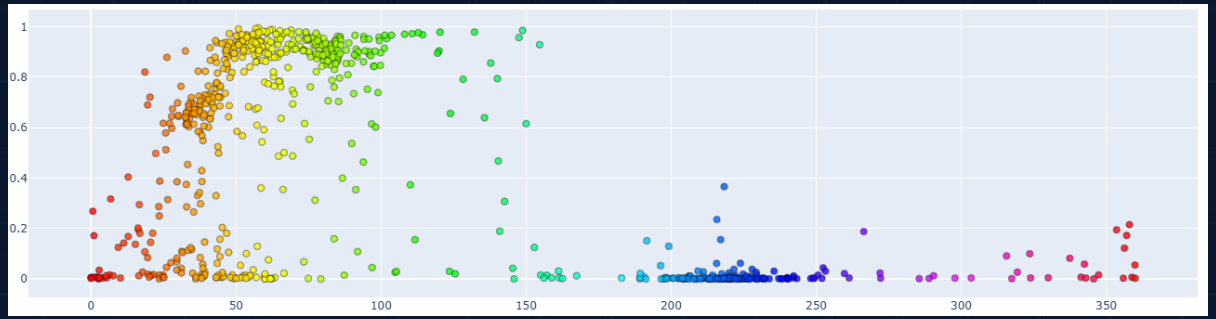


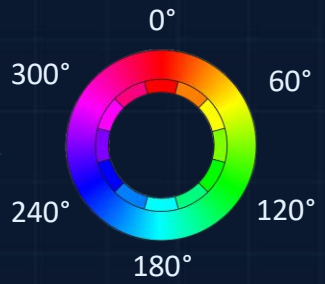
CNN Classification Analysis

Correlation between Mean HSV Values and Prediction Confidence



Probability %





Semantic Segmentation

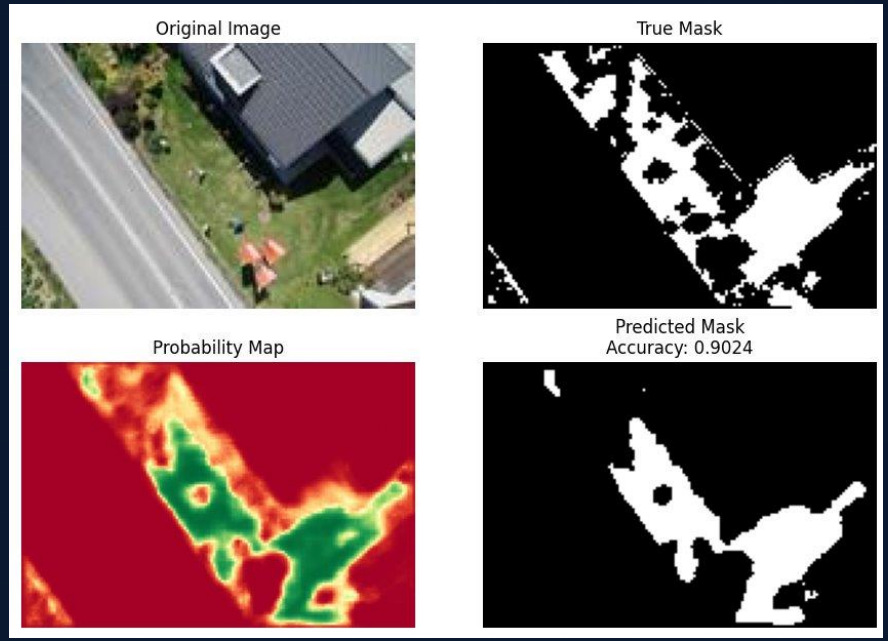
Semantic Segmentation

Problem Definition & Transition from Patch Classification

Pixel-Level Prediction

Unlike patch classification which assigns one label to a region, semantic segmentation predicts a class label for **every single pixel** in the image.

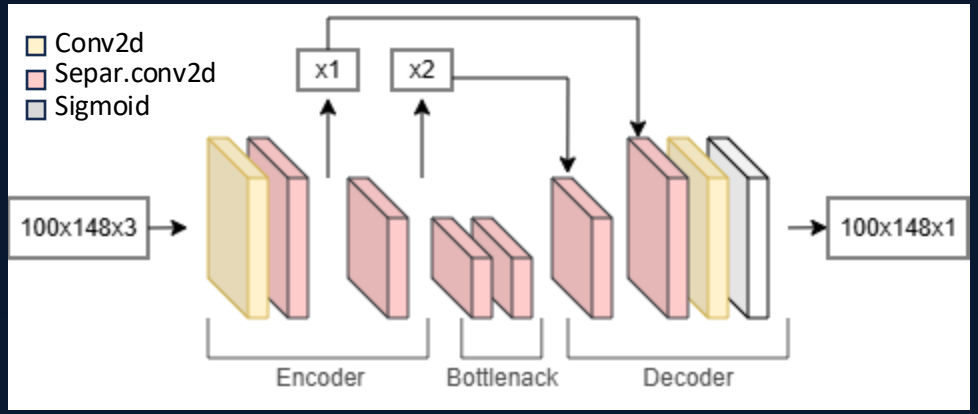
- Create a dense prediction map with spatial dimensions identical to the input.
- Requires understanding both local texture and global scene structure.
- Precise delineation of vegetation boundaries for accurate mapping.



To facilitate efficient training of the segmentation model, a specialized **reduced-size dataset** was created.

FCN Model Architecture

Encoder-Decoder Network for Dense Prediction



ENCODER

Downsampling Path:

Convolutional layers extract features while progressively reducing spatial dimensions. Captures "What" is in the image.

BOTTLENECK

Latent Representation:

- The most compressed layer containing high-level semantic information. Bridges the contraction and expansion paths.

DECODER

Upsampling Path:

- Reconstructs spatial resolution using upsampling or deconvolution. Recovers "Where" the objects are for pixel-wise output.

CNN pre-trained on patches for semantic segmentation

ADAPTATION LOGIC

We leverage the CNN model previously trained on **2,376 balanced patches**. Instead of training a segmentation network from scratch, we apply this robust patch classifier across the entire image using a sliding window approach.

EFFICIENCY GAINS

> Reduces training time significantly

> Utilizes learned feature representations

> Enables immediate inference on new data

Why does the Patch-Based Model predict "True" for non-vegetation pixels?
The model considers pixels as "True" (Vegetation) when they are located within a patch that is predominantly green. The internal features of the patch (e.g., 80% grass) override the specific characteristics of the center pixel, leading to false positives at the boundaries of vegetation areas.

Context Sensitivity vs. Pixel Precision

Final Model Comparison

Patch-Based CNN

Context Dependent:

Predictions are heavily influenced by the surrounding neighborhood of the pixel.

Spatial Averaging:

Learns to recognize the "patch" as a whole, smoothing out fine details.

Boundary Bleed:

Tendency to over-segment boundaries where vegetation dominates the patch area.

Accuracy: 72.88%

Semantic Segmentation

Pixel Precise:

Generates dense predictions for every individual pixel location.

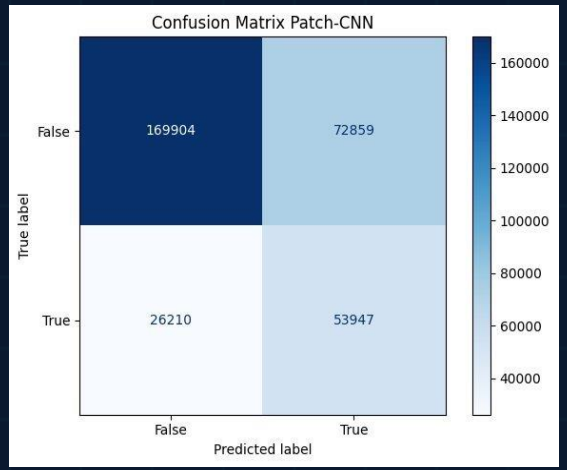
Boundary Aware:

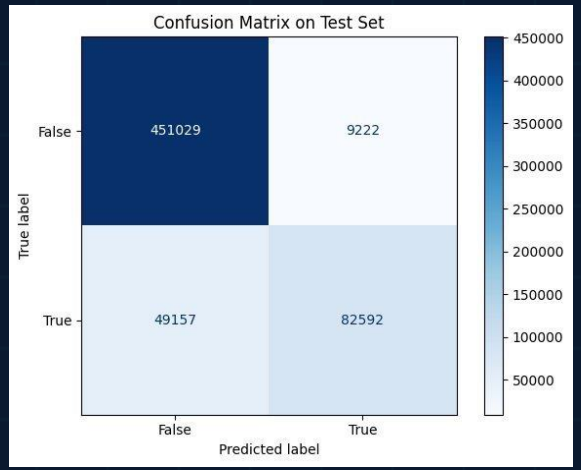
Preserves sharp edges between vegetation and non-vegetation regions.

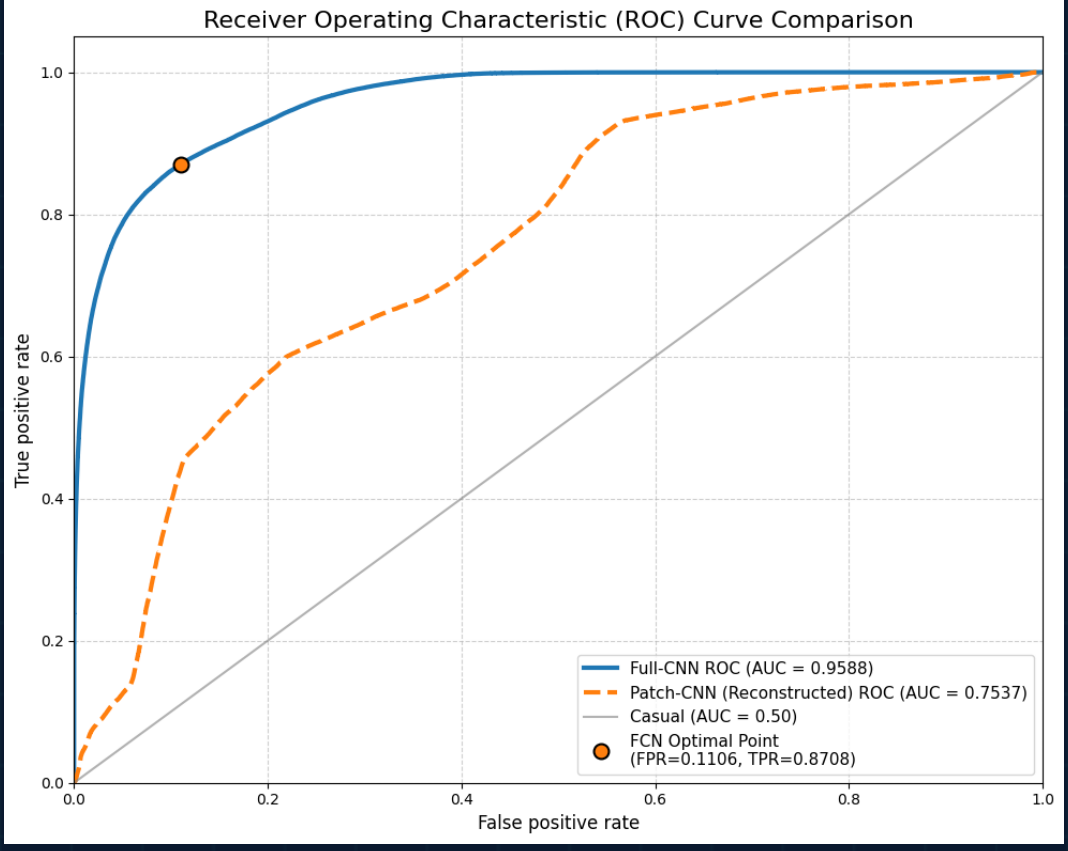
High Fidelity:

Captures small, isolated vegetation elements that patch-based models might miss or merge.

Accuracy: 89.10%





Model Comparison

ROC Curve